

# Hydrodynamic singularities

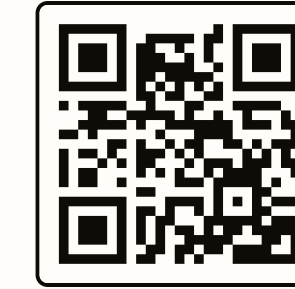
When a drop or a bubble pinches off, the flow forgets its past. A pinch of polymer brings the memory back — but only for the drop.

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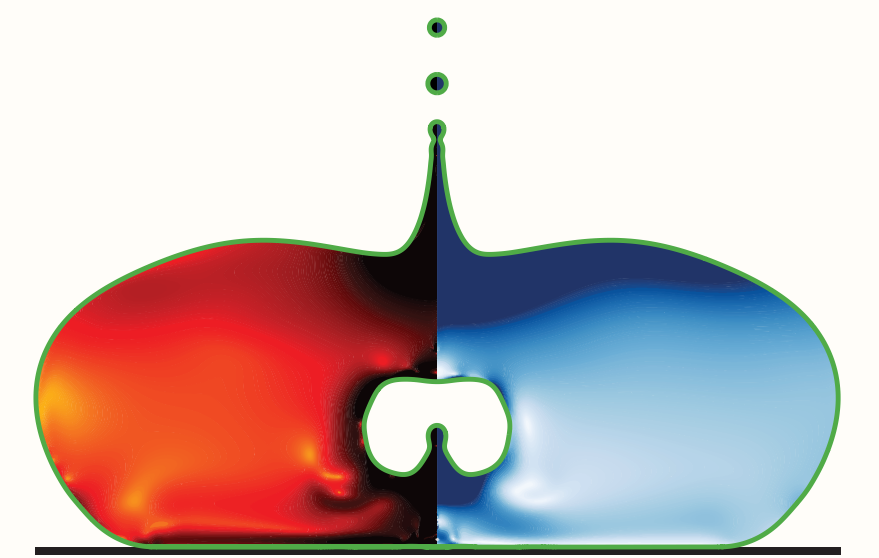
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Verschuur, C. I., Oratis, A. T.,  
Sanjay, V. & Snoeijer, J. H. How  
elasticity affects bubble pinch-  
off.  
arXiv:2511.20075 (2025)



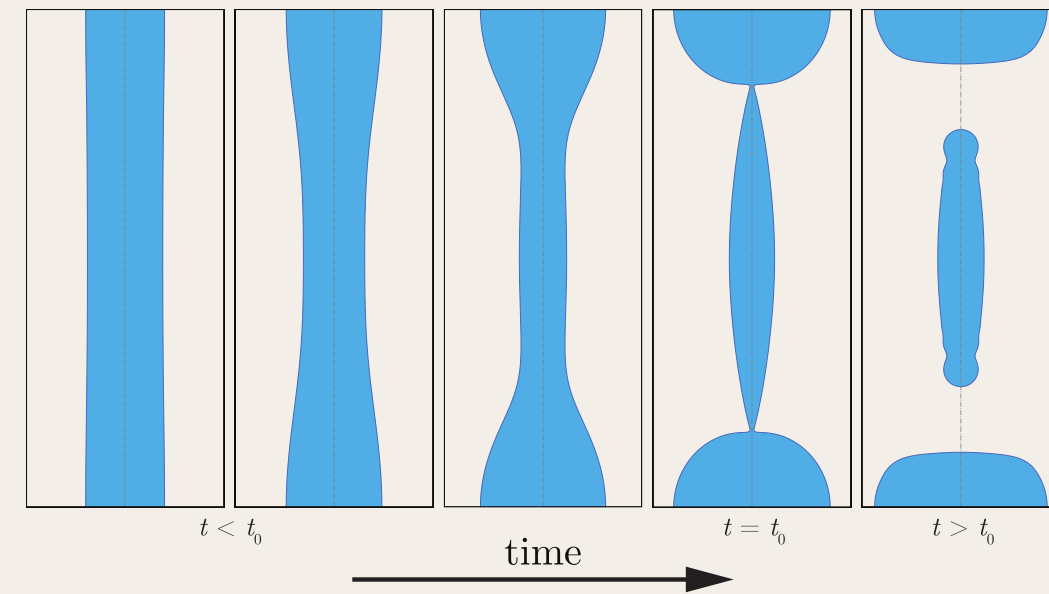
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A **singularity** is when a smooth physical process blows up — something finite becomes infinite in a finite time or at a finite place. As a drop or a bubble pinches off, the neck radius  $h$  races to zero at a finite pinch time  $t_0$ , and the shape turns self-similar in the time-to-singularity  $t_0 - t$ :  $h \sim (t_0 - t)^\alpha$ , a universal form set only by the local balance of inertia, surface tension and viscosity. The flow **forgets how it began**. Singularities erase a fluid's memory.

## A drop pinches off

$$h \sim (t_0 - t)^{2/3}$$

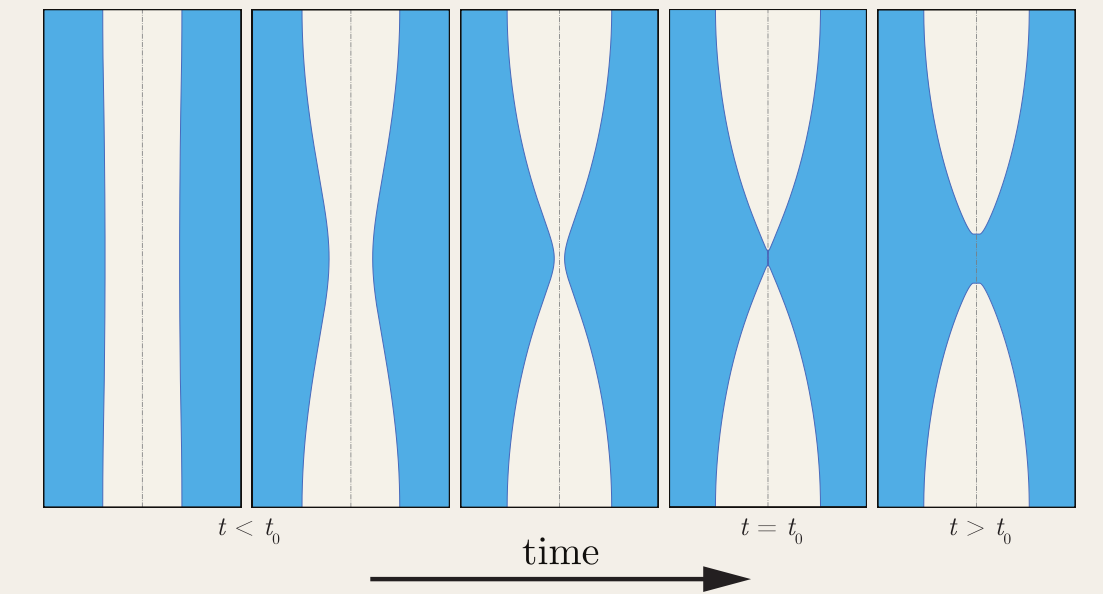
Plain water. Surface tension pulls a slender axial thread to a point, then it snaps into satellite drops — milliseconds.



## A bubble pinches off

$$h \sim (t_0 - t)^{1/2}$$

Plain water. The surrounding liquid's inertia collapses the cavity radially — microseconds, a thousand times faster.



Two free surfaces, the same finite-time blow-up: read left to right, the singularity is reached at  $t = t_0$ . They take different routes to it (a drop along a thread, a bubble by collapsing a cavity), so they carry different exponents in the time-to-singularity  $t_0 - t$ .

**Q** What if we gave the fluid a **memory** ?

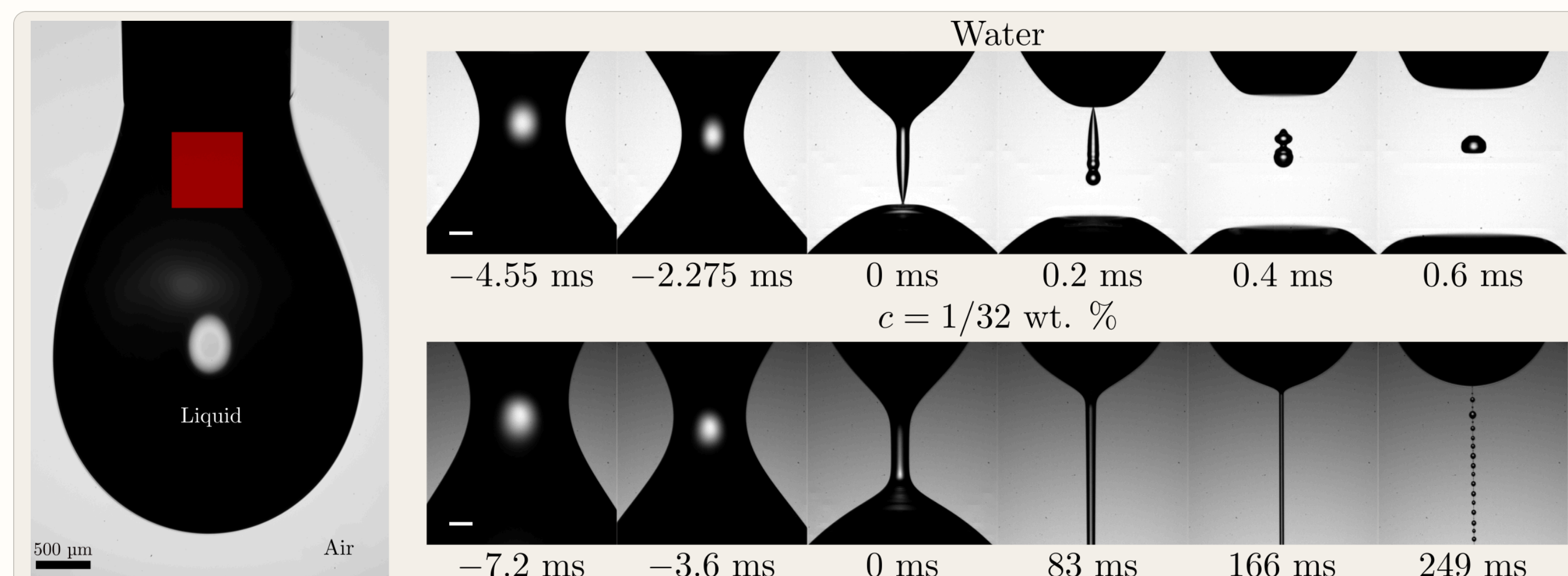
## Elasticity is memory

Dissolve a few polymer chains and the liquid gains a memory — each chain stores its stretching history over a relaxation time  $\lambda$ . Push to the perfectly elastic limit ( $\lambda \rightarrow \infty$ , nothing is forgotten) and ask whether memory can outlast the singularity. It does for a drop, yet a dilute bubble pinches off as if the polymer were not there. Why the difference?

## The drop remembers

A water drop reaches a finite-time singularity,  $h \sim (t_0 - t)^{2/3}$ . The pinch is capillary-driven, so the neck thins into a slender axial thread and the polymers stretch *along* it as it closes.

The elastic stress then diverges as  $\sigma_{zz} \sim G(h_0/h)^4$  — faster than the capillary drive  $\gamma\kappa \sim \gamma/h$  it fights. It must win: the collapse is arrested into a long-lived **beads-on-a-string** thread, and in the perfectly elastic limit the singularity is removed altogether.



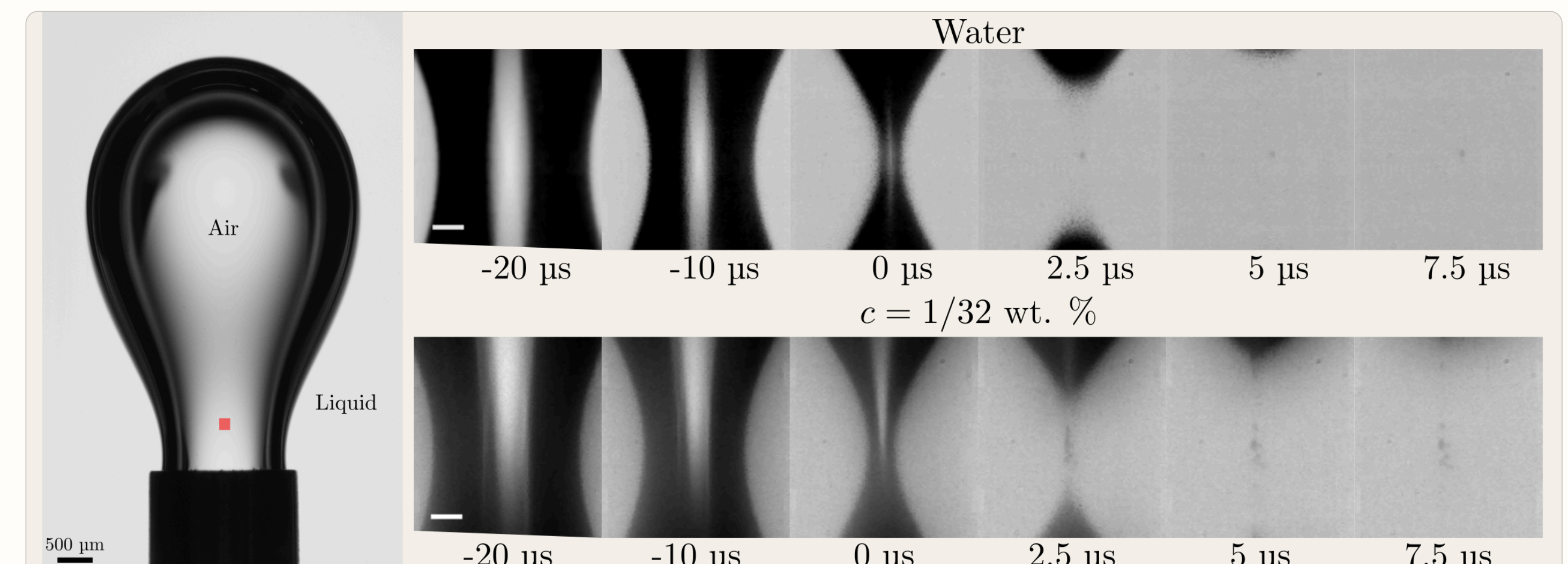
**Drop pinch-off.** Top: plain water reaches the singularity and snaps into satellites. Bottom: the same dilute polymer draws the neck into a persistent beads-on-a-string thread.

→ Dilute polymer → a thread. Memory wins.

## The bubble forgets anyway

A bubble neck pinches just as sharply,  $h \sim (t_0 - t)^{1/2}$ , but the collapse is driven by the liquid's inertia and the cavity closes *radially*, so the polymers stretch sideways rather than along a thread.

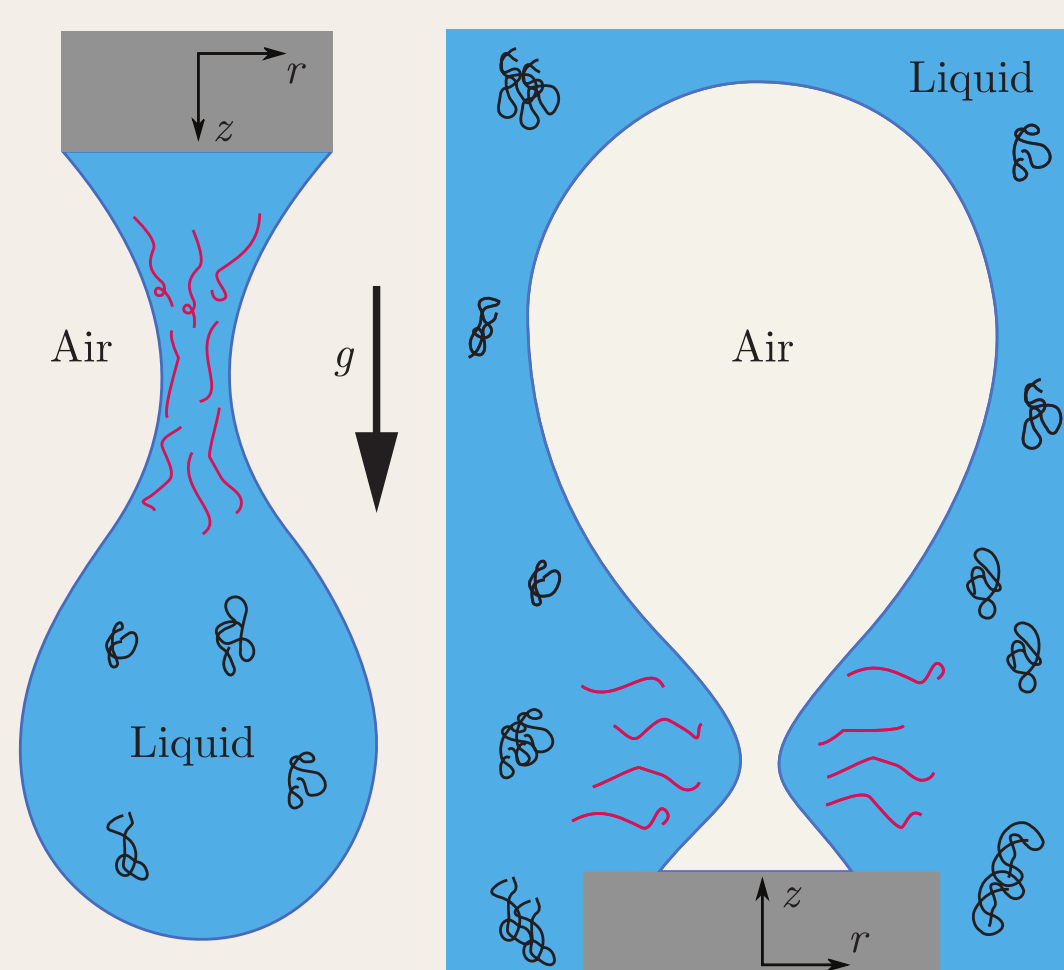
Now the elastic stress diverges only as  $\sigma_{rr} \sim G(h_0/h)^2$  — the *same* rate as the inertial drive  $\sigma_I \sim \rho\dot{h}^2$ . A fair race it cannot win: it stays subdominant, so the bubble reaches its singularity **with or without memory**.



**Bubble pinch-off.** Top: water. Bottom: the same dilute polymer. In the dilute limit the two near-singular sequences are almost indistinguishable.

→ Dilute polymer → no thread. The singularity survives.

## Why elasticity picks sides



Polymers (blue coils) stretch where the neck thins (red): **along** the drop's axial thread, but only **sideways** across the bubble's neck.

WHAT DRIVES THE PINCH

NECK GEOMETRY

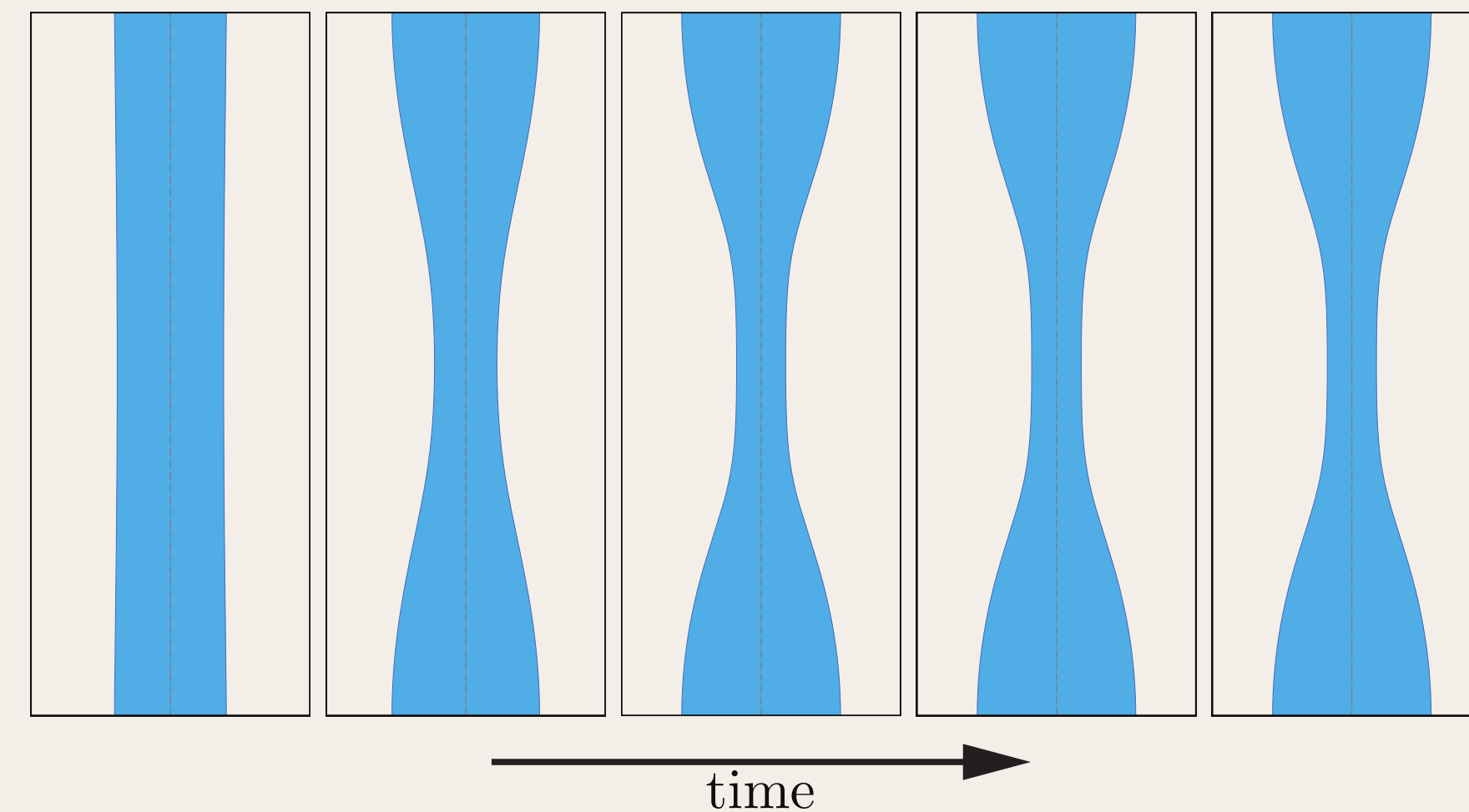
ELASTIC STRESS BUILDS AS

...VS THE DRIVE → FATE

The elastic stress in a drop outgrows its capillary drive, while that in a bubble just balances its inertial drive.

Same additive (polymers), contrasting fates — reading how elasticity rewrites singularities can teach us to control breakup, critical for inkjet printing, sprays and aerosols.

## Drop



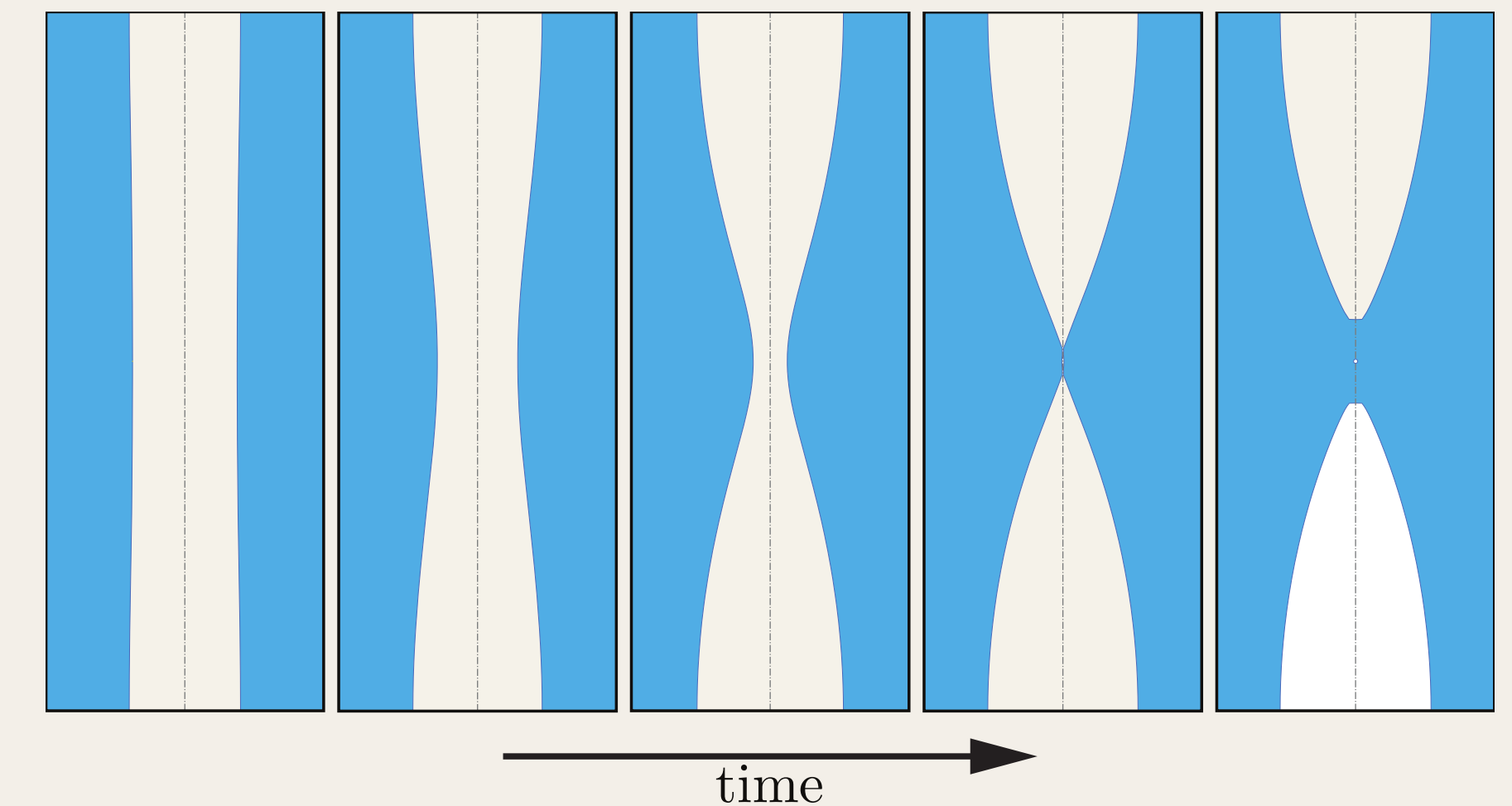
surface tension,  $\gamma\kappa \sim \gamma/h$

a slender **axial** thread

$$\sigma_{zz} \sim G(h_0/h)^4$$

**faster** (4 vs 1): a thread; singularity removed

## Bubble



the liquid's inertia,  $\sigma_I \sim \rho\dot{h}^2$

a **radial** cavity collapse

$$\sigma_{rr} \sim G(h_0/h)^2$$

**same** rate (2 vs 2): it pinches; singularity survives